"Statistical Analysis of Immunoglobin Isotype Fingerprints Across Afflictions"

"Statistisk Analyse af immunoglobin Isotype Fingeraftryk for Forskellige Sygdome"

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2026-05-05

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Introduction

Humoral Immune Response

Humoral immunity is part of the adaptive immune system and it is mainly mediated by the presence of antibodies. There are two main types of humoral immunity: T-dependent and T-independent. T-independent immunity is triggered by the presence of an antigen that is recognized by the B cell receptor (BCR), but not by T helper cells. This leads to a humoral immune response dominated by IgM antibodies. T-dependent immunity occurs when a B cell is able to internalize the antigen and present it to a T helper cell on the B cell's major histocompitability complex (MHC), allowing the B cell to undergo class-switch recombination to generate antibodies of other isotypes[1].

Antibodies

Antibodies are produced and secreted by B cells. They are proteins that form a part of the adaptive immune system. They play a central role in humoral immunity by attaching themselves to recognized targets to trigger various reactions. All human antibodies consist of a two identical light chains and two identical heavy chains. Both chains contain a variable region. The light chain contains a single conserved region, while the heavy chain contains either four (IgM and IgE) or three (the rest) conserved regions. This depends on which isotype of antibody it is. Between the first conserved region, which is connected to the variable region, and the second conserved region is a hinge region. Depending on the isotype, the hinge region can be longer or shorter, thereby controlling the flexibility of the antibody arms[2]. The different isotypes can interact with different groups of immune receptors, refferred to as Fc receptors.

The two arms of the antibody are referred to as fragment antigen binding domains (Fabs), and the bottom connecting part is refered to as the fragment crystallizable (Fc). The variable region of both Fabs is identical, as they are made from the same light- and heavy-chain combination. The hinge region is found between these two parts.

Variable Region

The variable region of an antibody compromises the N-terminal region of the light and heavy chains[3]. This is the part of the antibody that reacts with an antigen and contributes to activation of immune effector mechanisms[2]. The variability is generated through several mechanisms, allowing the body to produce antibodies that can rapidly become specific for a wide range of sequences or molecules that may appear.

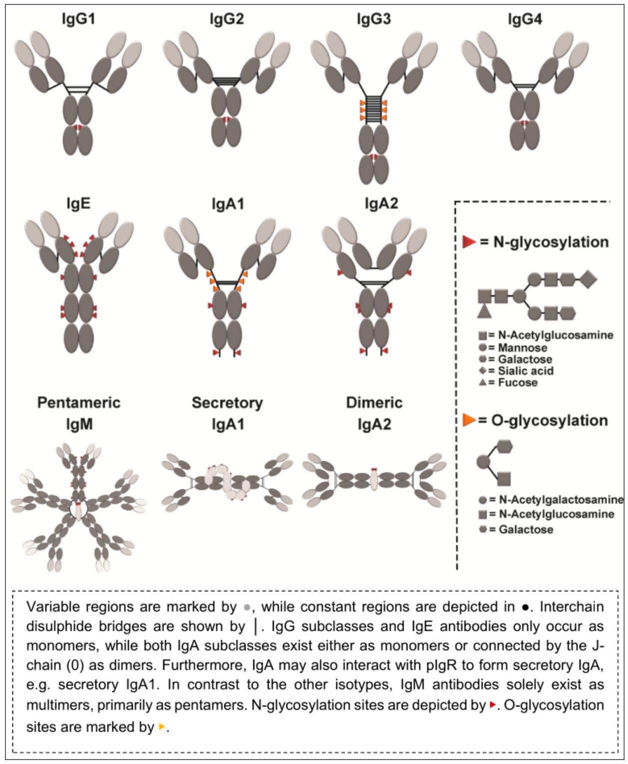

Figure 1: Graphic overview of antibody architecture by Kreschmer et al. 2017

Hinge Region

Between the Fabs and FC is the hinge region. The different isotypes and subtypes have different hinge regions. The hinge region of an antibody is found between the first and second constant regions of the heavy chain. The hinge region can be subdivided into three regions: upper hinge, core hinge, and lower hinge. Each of the three regions play a different role. The upper hinge determines the flexibility and rotational capacity of the Fabs. The core region contains a varying number of cysteine residues, which differs between subtypes. These cysteins are used to stabilize and connect the heavy chains. The lower hinge region is used to bend the Fc region of the antibody which affects its orientation relative to the antigen-binding arms[4]. The length and disulfide bonds of the hinge region affect the flexibility and reach of the variable region. They also affect Fc receptor binding affinity and downstream effector functions[2].

Constant Region

The main part of antibodies is made up of constant regions, these regions are stable, unlike the variable region. The light chain has a single constant region which is attached to a constant region of the heavy chain to hold them together. The constant region of the heavy chain is what determines the isotype of the antibody. The isotype of the antibody determines which effector function the antibody will trigger when binding with a target antigen[5]. The function of an antibody is dependent on which effector functions it triggers. The Fc region determines receptor specificity binding, thereby determining which immune receptors the antibody can cross-link with. Because the Fc region consists of the conserved part of the heavy chain, it determines which receptors an antibody can interacts with and therefore which reactions it can trigger. The antibodies and their respective receptors, function together to initiate downstream immune responses[2].

Variability

Antibodies are very variable proteins in terms of sequence. This variability can be seen in the variable region which can be adapted to consistently react only to very specific patterms. It is also expressed in the constant region which can be swapped out to mediate a completely different response than it would earlier. B cells are the producers of this variability, it is produced using several highly regulated mechanisms and processes inside the cell.

V(D)J Recombination

The variability of the variable region partially comes from V(D)J Recombination. V(D)J Recombination is different between the variable regions of the heavy chain and light chain. On the heavy chain it involves all three segments the V(ariable), D(diverse), and J(joining) segments, the light chain only has the V and J segments. By combining multiple different V and J segments it is possible to generate several different combinations of variable region to start mutating[3]. By combining these basic forms with inserted bases from SHM it is possible for the immune system to reach a high level of base diversity for affinity maturation to improve upon.

Activation-induced cytidine deaminase (AID)

One of the key enzymes in generating variability in antibodies is Activation-induced cytidine deaminase. It is a protein with a vital role in the adaptive immune system. It is used in the variability generating processes of CSR and SHM in order to induce double-stranded DNA breaks. The main mechanism of AID is the controlled deamination of cystein in the variable region of the antibodies. This turns the cystein into a uracil, which should not be in the DNA. When uracil is detected in DNA it triggers a doublestranded break in the DNA which is then repaired replacing uracil with cystein, as it originally was[3]. AID indirectly contribute to the DNA changing from break to break. When a break is induced in DNA it is normally repaired with very high precision. During a breakage caused by AID the normal DNA repair pathways are suppressed, which causes the cell to use less reliable means of reparation. These less than ideal means cause the DNA to be imperfectly repaired, leading to mutations in the variable region where the AID is allowed to act. By inducing mutations in this region of the DNA it increases the diversity of variable regions in this B cell lineage[6].

Somatic Hyper Mutation (SHM) Mutations happen to the variable region in B cells over the generations of .Somatic hyper mutation is how the B cell performs random insertion, deletion, and base swap mutations between generations. It mainly occurs in the variable region of the DNA where the AID is supposed to be acting. This causes a buildup of minor changes in the DNA making up the variable region, thereby increasing the number of potential combinations that can be created[3]. Higher affinity mutations are selected for further mitosis, this causes the next generation of antibody producing B cells to produce antibodies of higher affinity[7].

Affinity Maturation These higher affinity B cells are chosen through competition. The overall process is referred to as Affinity Maturation. Affinity maturation is one of the main mechanisms of the adaptive immune systems adaptation to diseases. It uses SHM to rapidly generate a diverse range of variable regions, and then select the best ones for further production and adaptations. B cells that have been activated, cycle between two zones: the dark zone (DZ) and the light zone(LZ). In the light zone B cells compete for the attention from T folicular helper cells, which select the highest affinity cells, to enter the dark zone to replicate. Inside the dark zone B cells divide and SHM triggers more mutations, most of these mutations will have a negative impact on the affinity of the B cell. After a few rounds of replication, the B cells are then pushed out of the dark zone into the light, to once again fight for the right to to enter the dark zone for replication[8]. While the affinity maturation cycle is running, the B cells are also exposed to evironmental factors like cytokines that can induce class switch in them.

Class Switch Recombination (CSR) When certain criteria are meet the B cell will edit its own DNA and switch from one class to the next. Class switching is a one way process. Class switch recombination is the process by which B cells change the constant region of the heavy chain of the antibodies they produce. This is done by splicing different parts of the DNA together, cutting out the middle from the splicing. This makes it impossible for the transformation to go backwards, as the DNA is not there. Even though there is no mutation during CSR, AID is necessary to trigger the double strand breaks to excise the DNA before recombining the DNA. The excision points are predefined by switch regions in the DNA. These regions are used as target points for the DNA break to ensure that the breakage is clean and that there are no problems with the new heavy chains[9]. With the excision of only some DNA it allows the downstream B cell generations to perform further CSRs for further adaptation[8]. Which DNA is excised depends on which cytokines are present at the time of activation, which decides the isotype that the B cell will change to.

Fc Glycosylation

Post-translational modifications (PTMs) play an important role in the function and behaviour of proteins. Different isotypes have different patterns for post-translational modifications. It is one of the things that differentiate the functions of the different antibody isotypes. By switching isotype the potential modifications available for the antibody also change. And due to a positional change the same type of PTM can now have a different effect on the interactions of the antibody.

Potential modifications include glycosylation, which is of special importance to the function of antibodies. Glycosylation is the attachment of a oligosaccharide molecule to an amino acid. It can occur in two overarching forms: O-glycosylation where the oligosaccharide is attached to the oxygen atom of the hydroxyl group of a serine or threonine and N-glycosylation where it is attached to the nitrogen atom of an asparagine side chain. There are several different types of glycosylation, named after the oligosaccharide that is being used. These different oligosaccharides have different effects on the function of the protein they are bound to. The oligosaccharide used has also been observed to vary from continent to continent, whether this is due to genetics, diet, or other circumstances is unknown. But it is known that glycosylation type can change with age, and that indicates that it is potentially a combination of several different factors[10].

Glycosylation is one of the main differentiators in functionallity between the isotypes, and their subtypes. The different isotypes have different amounts of asparagines that can accept N-glycosylation. Glycosylation and its effect on antibodies has mainly been studied for IgG antibodies. N-glycosylation of IgG Fc region mainly happens in the cells where the antibodies are produced. The glycosylation of Asn297 in IgG opens the configuration of it Fc domain. Antibodies are not the only part of the humoral immunity that is affected by glycosylation. FcγRIIIa has five N-glycosylation sites, the IgG Asn297 glycosylation site can interact with the Asn162 site of the immune receptor, to promote antibody receptor cross-linking. Glycosylation status is not a permanent thing, and it is not absolute. Only 12% of IgG3 are glycosylated on Asn392 which it alone amongst the IgGs possess. Glycosylation also directly affect the functionallity of the antibodies themselves. Glycosylation in IgG has been linked to increased binding of antibodies to C1q, which activates the classical complement pathway[10]. It plays a vital role in combating diseases and that has been noted by some pathogens.

Some diseases affect the body's ability to perform different glycosylations on certain antibodies. This generally occurs by lowering the amount of enzymes that are available to perform that specific type of glycosylation. This can have what appears to be a targeted effect on certain antibody isotypes. It is possible to use changes in IgG glycosylation of Asn297 to find out if someone has active or latent tuberculosis. evolutionarily this is probably a survival mechanism of the pathogen. It also indicates that the glycosylation plays an active role in the function of the antibodies[10]. With glycosylation being so important the differences between isotypes in terms of glycosylation is a part of the ability that separate the different isotypes.

Isotypes

Antibodies come in five main isotypes, some isotypes have several subtypes. The different isotypes differ structually in various ways, not just in their length but also in their glycosylation and their cystein placement. It is these structural differences that the body uses to selectively interact with the individual antibodies to mediate which reaction is triggered by which isotype[2]. Although there are many different, at the root of it all is the same isotype.

IgM

IgM is the default isotype of mature B cells. It contains 4 conserved regions, and in the blood they form pentamers. They are the best activators of the complement system. IgM can be produced without being exposed to antigenic stimuli, but they can also be produced against invading pathogens[2]. It serves as the primary defense against early pathogen invasions, before adaptive immunity has been activated[3].

Early Immune resposes are generally characterised by a rise in antigen-specific IgM, which is then followed by affinity maturation, and then by isotype switching. This gives rise to antibodies that are specialised in combating that specific illness, therby increasing the likelyhood of survival for the host. IgM tends to have a lower antigen-binding affinity than the other isotypes, in nature this is compensated for by being the only antibody to naturally be able to target molecules other than proteins[11]. This is because in order to class-switch it is necessary for the B cell to react to an antigen that can be recognized by the MHC II receptor on the folicular T-cell, which can only recognize proteins. Clinically humans have overcome this by attatching proteins to molecules of interest, thereby allowing the Tfh-cells to induce class switch on B cells that are not protein specific[1]

IgD

IgD is the other type of immunoglobin that is produced by naive B cells[5]. It is believed to have arisen evolutionarily slightly after IgM, and is preserved from fish to humans. This kind of preservation would indicate that it is necessary for the survival of the organism. The maturation of B cells is characterized by progressive up-regulation of IgD based receptors. B cells that purely express IgD are anergic B cells and tend to exist in the perepheri parts of the body, but interestingly not in the intestine[12]. This may be due to the presence of the intestinal microfauna, but if IgD were a danger to the commensal microbes there should be a similar situation for IgA antibodies, which are the main excretory antibodies.

IgD stands out amongst the isotypes for several reasons. One of the more noteworthy ones is that it can coexist with another isotype. This is because it is an alternative splicing of the same gene that produces IgM, but unlike other subtypes it has an individual gene to code for its heavy chain. Although it initially coexists with IgM, it is possible for B cells to class switch so that they only express IgD. The process of CSR to IgD appears to differ from normal CSR, which is another reason to think that IgD should be considered separate from the other isotypes[13].

With IgD being an alternative splicing of IgM there is an argument to be made that it should really be called IgM2 and be classified as a subtype of IgM. Some arguments against that would be the existance of an individual gene for its heavy chain[13] and unlike other subtypes, that seem to be specialised versions of each other, IgD fills a distinctive role in the immune system in terms of functions due to having independent receptor reactions from IgM[12].

IgG

IgG antibodies come in four subtypes, IgG1, IgG2, IgG3, and IgG4. These subtypes are numberd in order of decreasing abundance. All four subtypes can be transcribed by the G class B cells. The different subtypes have different affinity for different receptors of the Fcγ family. Through the different affinities they are able to fulfil different purposes in the immune system. With the exception of FcγRIIb, all of the FcγR receptors mediate activating functions[14].

IgG1 IgG1 is the bodys response to soluble proteins and membrane proteins. They are the most abundant of the IgG antibodies, and are often encountered together with the other IgG subtypes, especially IgG3 and IgG4[15]. IgG1 is able to cross-link with all of the different FcγR receptors to activate them[14].

IgG2 IgGs reaction to bacterial capsular polysaccharides is almost entirely made up of IgG2 antibodies[15]. IgG2 mainly triggers receptors on neutrophiles, it might also be able to trigger receptors on NK cells and some macrophages[14]. This would lower the scope of the reaction, triggering only the necessary cells for fighting the bacterial infection. Therefore no antibodies are wasted on other activities, leading to an improved immune response.

IgG3 IgG3 is the antibody with the longest hinge domain, counting up to 62 amino acids in length. It also has the highest affinity for C1q which makes it the best IgG to trigger the complement system. It does have the shortest halflife amongst the long lived IgGs. Just like IgG1 it can interact with all of the FcγR[14].

IgG4 IgG4 is an antibody that is often formed following repeated non-infectious exposure. It is linked to downregulation of immune reaction towards the antigen and is usually seen correlating with sucessful immunotherapy[15]. IgG4 has a couple of unique functions as compared to the other IgG subtypes. It doesn't have a C1q domain for triggering the complement system[16]. It also has the ability to exhange half of its molecules with other IgG4 antibodies, which would allow 1 antibody to target two separate patterns for immune suppression[17].

IgA

IgA is the most produced antibody isotype in the body, it is the second most prominent antibody in the blood. More IgA is produced in total than all the other isotypes combined. It is encountered in all mammals and birds. Most IgA is secreted out of the body, usually in the form of SIgA, which mainly forms dimers outside of the body, and sometimes they also form tetramers. IgA comes in two major subtypes, IgA1 and IgA2[18].

There are several differences between the two subtypes. The two main differences between the two subtypes, are the length of their hinge regions and their glycosylation sites. The significantly longer hinge region of IgA1 makes the fabs with the variable regions significantly more flexible, and give them a longer reach. In the serum there is roughly a nine to one ratio of IgA1 to IgA2 while in the mucosal tissue the ratio is less lopsided. The two subtypes induce different signalling in some cell which is probably due to a combination of the hinges being different and the differing glycosylation patterns. The production of two different subtypes of IgA is something that is only found in the ape family from which humans sprung out[19].

IgE

IgE is the antibody type mainly associated with allergic and parasitic diseases. It mainly interacts with receptors on basophils, monocytes and macrophages, with some other immune cells carrying lower affinity receptors. When an antibody-receptor complex is triggered the triggered cells degranulate completely, this degranulatinon triggers high levels of inflammation and cascading reactions[2]. The relatively higher affinity of IgE receptors, as compared to other antibody-receptor complexes, drastically lowers the amount of IgE that is freely found in blood serum.

IgE is the least abundant antibody in the blood. This is probably partially due to the lower life lifespan when unbound, which is 1.5 days[12], and the fact that it is produced in response to parasites, which are more rare than microbial lifeforms. In some individuals IgE antibodies are produced as a response to harmless antigens, this failure can lead to the symptoms of allergy[6]. Perhaps the growing prevalence of allergy is due to the lower variation in IgE leading to an increased amount of Mast cells being triggered by a single harmless antigen.

BCR

B cells don't just make antibodies for other cells to use, the B cells themselves can use the antibodies to build a receptor. The B cell receptor is a membrane bound receptor that is constructed using antibodies. Immature antibodies use IgM as the antibody for the BCR, but as they mature it is slowly swapped for IgD. All of the isotypes can be used for constructing BCRs. The isotype used is based on the class switch of the B cell. By using different isotypes it is possible for the B cells to have different reactions to the same receptor being triggered. An antibody that forms a part of the BCR is referred to as a membrane bound immunoglobin, or mIg for short. On the Fc end of the antibody is attatched a trans membranal anchor section, which anchors the antibody through the membrane and acts as an intercellular effector when the antibody is triggered. The effector is associated with CD79A and CD79B which each carry an immunoreceptor tyrosine-based activation motif (ITAM). These Motives are what allows the BCR receptor pass on the signal[20].

There are several theories about how the BCR receptor works, but it always seems to have some behaviour that defies the logic of the model. While the model may not have been established, the basic sequence of events is known. The process starts with a variable region of the BCR. It is either triggered by a random floating fragment, or more likely, deliberately triggered by antigen presentation by folicular dendritic cells. The BCR reacts by triggering the ITAM part of CD79A and B. The ITAMs then trigger a cascade of reactions that always lead to transcriptional changes in the B cell. Which transcriptional changes occur depends on the isotype used in the receptor[20].

Effector functions

When BCR, or any other antibody is triggered a cascade of reactions termed effector functions follow. Antibodies have the ability to trigger two main types of effector function, either they bind to receptors to trigger transcriptional change, or they activate the complement system. which reaction occurs depends on the antibody isotype, based on if they have a C1q domain and which Fc receptors they can bind to.

Complement System

The complement system is one of the basic defence mechanisms of the body. It consists of a series of proteins, that through a proteolytic cascade of reactions lead to opsonization, recruitment and membranelysis. A complement activation is a three phase process, initiation, activation and effector phases. The initiation phase of the complement system has three pathways, the classical, alternative, and lectin pathways. The classical pathway is initiated by antigen-antibody immune complexes. It is activated when an antibody with a C1q domain binds to something. The ability of an antibody to activate the classical pathway is directly related to the affinity of their C1q domains. Which determines the ability of the antibody to bind complement proteins. For IgG the C1q affinity is ordered IgG3 > IgG1 > IgG4 > IgG2. As the complement cascade progresses, a part of the proteolysed complement proteins have an inflammatory effect on the immune system. If the target isnøt removed in other ways, the complement proteins form MAC the Membrane Attack Complex which lyses the victim[21]. For IgG the C1q affinity is ordered IgG3 > IgG1 > IgG4 > IgG2. As the complement cascade progresses, a part of the proteolysed complement proteins have an inflammatory effect on the immune system. As the cascade progresses, if the target hasn't been removed in other ways, the complement proteins come together to form MAC, the Membrane Attack Complex, which lyses the victim[21]. The complement system plays a part in both innate and adaptive immunity, if the pathgens do not succumb to the hostile environment of the body, the cells start to remove them.

Immune Receptors

The body has several immune receptors that play different roles in the immune system. A subset of these receptors can interact with the Fc of antibodies to trigger various effector functions, such as opsonization, and transcriptional change. If the antibody is attached to an immune receptor when it binds to something it will trigger the immune receptor, and it is still possible for the immune receptor to be attached after the antibody has found its target to trigger it. There are several types of immune receptor, as it is a categorisation for all receptors involved in the immune system. A subset of these receptors work by binding to the Fc of antibodies to cause an effect, these are the Fc receptors.These Fc receptors help coordinate the humoral immune response through the usage of immunereceptor tyrosine-based activation motif. The motives are triggered when the receptor is cross-linked with an antibody. These receptors can trigger a wide variety of downstream effects, that can vary depending on the cell type of the receptor holder[22]. Immune receptors naming follows a simple convention, where Fc_R_ stands for "Fragment crystallizable *type* Receptor *number*". An example is FcγRII, which is Immunoglobin G receptor type 2[2]. What reactions occur depend on the isotype of the antibody, and that depends on which disease caused the class switch.

Diseases of Interest

The ideal trigger B cell related activity are pathogens. These pathogens will hopefully be recognized by the B cells and then be a target of specific eradication with the help of affinity maturation and class switching. This will change the population distribution of antibodies based on the type of pathogen that is present. The transcriptional changes induced as a reaction to the disease will affect which effector functions are used against the pathogen. The different pathogens require different solutions, and the immune system reacts based on the identitiy of the pathogen.

Viral One common type of pathogen is the virus. It is a non-living bundle of DNA protected by a coating, with the goal of hijacking the replicative machinery of a living organism to create more copies of itself. A common type of viral disease is Viral Respiratory Infection. It is the most common category of infectious disease, and a leading cause of death worldwide. With a growing population and climate change the threat of viral diseases is increasing yearly[23].

Bacterial Another common pathogen is bacteria. Bacteria have a long evolutionary history, and have therefore had plenty of time to adapt to various conditions. They occupy a broad variety of ecological niches, so it is not wierd that there are some of them that have humans as their niche[24].

A gruesom category of bacterial diseases is bloodstream infections. When bacteria starts colonising the blood vessels and breaking down the body. These bacteria will rapidly colonize the blood vessels, while doing what they can to avoid the immune system.

Fungal

Parasitic

Allergies

Auto-immunity - ADD CITATION FOR B CELL DEVELOPMENT In autoimmune disorders the adaptive immune system targets the body itself. In the case of antibodies this can be categorised based on the isotype of the antibodies involved. Antibody mediated auto-immunity is related to the variable region of the antibody. Normally auto reactive antibodies are sorted out and removed during early B cell development. This protects the body from itself and hinders the development of antibodies that recognize the host organism. In auto-immunity some B cells have developed that recognize host antigens, and have not been removed.

IgG mediated auto-immunity Not much research has been done on auto-immunity in the context of antibody isotypes, outside of IgG. IgG is the most abundant antibody in the blood, ant it is also the most routinely measured. This has lead to a wealth of data from auto-immune patients in terms of its presence and distribution. In terms of IgG mediated auto-immunity it would appear that IgG4 plays an important role. IgG4 is usually associated with downregulation of immune reactions, and has the number 4 becuase it is normally the IgG antibody with the lowest concentration level in the blood. In data analysed by Volkov et al. in 2022[17] many cases of IgG mediated auto immunity had a very high level of IgG4 antibodies, which could implicate its immune suppressing capabilities in auto immunity.

IgG4's role in autoimmunity is so common that the IgG4-related disease (IgG4-RD) is an area of study. Hightened IgG4 often appears together with other isotypes that cause autoimmunity[16]. This could be related to IgG4 being suppressive, so it would target antigens that should not cause an immune reaction.

Proteomics

Theory says that different antibodies are produced against different disease. Proteomics is how theories are proven - or disproven - in the field of protein research. Proteomics is the study of a large amount of proteins from the same source. It allows for the answering of questions in relation to the abundance, localisation, interactions, modification states, and folding of proteins. Proteiomics comes in two main types, bottom-up and top-down. In bottom-up proteomics an overall picture of the protein abundance is created and then further studied in order to get an overviw of the situation, and find out how the different proteins play together. Top-down proteomics on the other hand makes it possible to analyse isoforms of the same proteins, but it becomes harder to gain a complete overview[25]. There exist many tools for performing proteomics, but modern proteomics is mainly done on a mass spectrometer.

Mass Spectrometry

Mass spectrometry is performed using a mass spectrometer. A mass spectrometer is a moleculart weight that is capable of weighing the individual molecules of a sample. A mass specrometer is made up of six components, an inlet system, an Ion source, a mass analyser, a Detector, a vacuum system, and a recorder. Of these six components, three (ion source, mass analyser, and detector) can be varied for different types of analysis. These three components work together to first ionise the sample, sort it, and detect the molecular composition and concentration distribution of the sample.

A mass spectrometer starts with an ion source. The purpose of the ion source is to ionises the sample molecules. The ionised particles can then be converted from a non-gaseous phase to a gaseous phase so that they can be analysed by the mass analyser. There are several ways to ionise a molecule, and that has lead to the creation of several different ion source instruments. Ionization is split into hard ionization and soft ionization. Which to use depends on once more on the molecules that are being analysed. The categorization of hard and soft corresponds to high and low internal energy being transfered to the sample molecules during the ionization process. For biological samples with complex molecules, soft methods are preferable. They do not break the molecule in the process which would make the analysis unreliable[26]. Currently the two main ion sources used in proteomics are matrix-assisted laser desorption/ionization (MALDI) and electrospray-ionization, which are both soft methods.

The mass analyser is the part of the mass spectrometer that is used to affect the ions so that the detector can detect differences between them. The sample molecules are separated based on their mass to charge ratio. Mass analysers uses several different methods to separate the ions based on their characteristics. Currently there are four main types of mass analyser in use, quadrupole, quatrupole ion trap, time of flight, and fourier transform ion cyclotrone resonance. The different mass analysers are used for different purposes based on several parameters such as resolution and price of usage. In order to analyze only particles within a certain m/z range several mass analysers can be put in sequence, this is referrred to as tandem mass spectrometry[27].

After the mass separation a relevant mass detector is used to determine the mass of the ion based on how it has been inside the mass analyser. Mass detectors are analytical devices which detect the passing or incident of ions and their absolute or relative concentration of each analyte in the sample. Over the history of mass spectromtry, there have been several different types of mass detector. Many of these mass detectors are used in conjuction with specific mass analysers. When a detector is triggered it outputs a signal, that after amplification is traced on a recorder[28].

Bottom-up Proteomics There are two major categories of mass spectrometry in the field of proteomics, Top-down proteomics (TDP), and Bottom-up proteomics (BUP). Bottom-up proteomics is the analysis of proteins that have been proteolysed by enzymes[29]. The goal of of BUP is the efficient, consistent, and effective extraction and digestion of proteome samples for analysis. The general workflow of BUP is made up of; protein extraction, followed by hydrolysis, Reversed-phase liquid chromatography (RPLC), ionization, and then finally mass spectrometry. Depending on circumstances, various changes can be made to this flow to fit whatever criteria that are necessary[30].

Top-down Proteomics The major difference between BUP and TDP is whether the proteome of interest in predigested or not. The predigestion changes the difficulty of answering different questions. The digestion of proteins that is characteristic of BUP has the side effect of making it hard to analyse the post-translational modifications of the proteins, which are important for their functions. There are of course also challenges to using TDP, some of them are the need for both accurate intact molecular mass measurement(top) and controlled fragmentation during the gas-phase[29].

DDA Mass Spectrometry Data-dependent acquisition (DDA) is a form of tandem mass spectrometry. It functions by selecting the most abundant precursor ions detected by MS1 survey and then scan them after fragmentation[31]. One of the problems with DDA is that it has a bias towards stronger signals, that makes it harder to reproduce quantification of low-abundance peptides[32]. One attempt to solve this issue was the usage of dynamic exclusion settings, which mitigates the problem to some extent. But this is an inherent flaw with DDA and has lead to the creation of new techniques like DIA mass spectrometry[31].

DIA Mass Spectrometry Data-independent acquisition (DIA) is the newer approach to data acquisition[32]. It tries to solve the bias issue of DDA by using a different approach. DIA allows for the isolation and fragmentation of all the acquired precursors, by cycling through predefined *m/z* windows across a preset range[31]. This eliminates the bias by making each precursor the largest in its cycle and therefore allowing the identical treatment for the different precursors regardless of abundance.

Labeling When analysing samples it is important to know if and how the samples are labeled. Labeling samples makes it possible to run them together. This decreases the potential for technically induced bias caused by running the samples seperately. It does also add the extra step of labeling the samples, and if it isn't done correctly that can also affect the outcome of the experiment. Two types of labeling are; stable isotope labeling, and isobaric tag labeling[30].

Stable isotope labeling rely on feeding the sample with stable isotopes of the atoms it uses, generally carbon. This produces peptides that are chemically identical while having a slightly different mass[30]. The obvious downside of this method is the labeling process itself, it works for microbes, but is probably impossible to perform at scale with anything larger. And the isotopes are expensive to produce and procure.

Isobaric tag labeling is another approach to running several samples at once. There are various methods of performing isobaric tag labeling. The basic approach is to attach a different isotopic mass tag to all of the molecules in a sample. These mass tags can then later be used to differentiate between the samples. The tags themselves are stable isotope reagents that can be attached to to the peptides, and differ enough that it is possible to note the difference between them during analysis[33].

Datasource

PRIDE is the online protein repository of the European Molecular Biology Laboratory (EMBL). It stands for The PRoteomics IDEntifications Archive database, and is a public data repository for mass spectrometry data. The goal of PRIDE is to support the performace of reproducible research through the application of quality control metrics and enable the reuse of data for purposes other than the original study. The source of the datasets in this study is PRIDE and each dataset contains the ftp address from which it came[34]. PRIDE has four major web interfaces: PRIDE archive, PRIDE Archive USI, the PRIDE Crosslinking resource, and PRIDE spectral libraries. These different resources fulfil different functions in the PRIDE ecosystem. PRIDE supports the FAIR (Findable, Accessible, Interoperable, Reusable) principals, and is constantly working towards the furtherance of these concepts in the database[35].

FragPipe

FragPipe is a application designed to be a comprehensive computational platform for the analysis of mass spectrometry proteomics data. It functions as a wrapper over several proteomics tools, such as MSFragger, IonQuant, DiaNN, and more. It provides a comprehensive list of workflows that combine the tools to turn multistep processes into a set up and run. The analysed data is generally turned into tables of quantified proteins, which are easy to work with for future analysis. FragPipe is developed and maintained by Nesvilab. it is open source, and under continuous development. FragPipe documentation can be found at the url https://fragpipe.nesvilab.org/, which is made with github pages and is sourced from the https://github.com/Nesvilab/FragPipe.git github repo[36].

DIA-NN

Data-independent acquisition Neural Network, is the main tool used by FragPipe to analyse DIA data. It uses deep neural networks to solve one of the largets issues with DIA mass spectrometry, the analysis of the data. As data is not aquired in one round, but rather through cycling it loses the context window that makes it easy to analyse the data. With no baseline DIA data is hard to analyse and peaks can be confused for several precursors at once, as there are several corresponding spectrums. The neural networks are used to distinguish real signals from noise, it combines those predictive capacities with at the time of creation new quantification and interference-correction methods[37].

When it starts it uses a selection of precursor ions to build a spectral library which is used as a foundation for future quantification of the peaks. It then generates a negative control library from the decoy precursor information, and uses that to identify potential peaks that should overlap between the decoy and the real data. After the preparatory work it then procedes to evaluate every elution peak with a set of scores that describes the peaks characteristics, while comparing with the generated spectral library. It then evaluates the degree of interference between the precursors with similar retention time, and if the interference is significant enough to take into account. Then it keeps the ones that have the least interference, this has been found to increase the strict false discovery rate (FDR). One of the most important steps in the workflow is that it assigns statistical significance to the identified precursors, in the form of q-values. The q-values are calculated by assigning a discriminant score based on the potential characterisations of the elution peaks. The program then uses the trained neural network to distingush between the decoys and the actual peaks. It then proceeds to select the most likely fragment based on the level of interference and best correlation[37].

SDRF-proteomics - Needs to be worded better

After data has been gathered and analysed, it is important to store it for future analysis. As evermore data is generated, it is becoming increasingly important to improve and standardise the presentation of this data. With the evergrowing volumes of data that are being generated this is becomming more and more important. But what data is actually important for the understanding, and reproduction of the experiment. There are two types of data that are important: the data that makes it possible to understand the experiment, metadata, and the data that tells describes the process and makes it possible to reanalyse the data. These are two separate goals, and will need two separate solutions. The first is solved by using a meatadata file which contains the names of the columns in the data, what they represent, and maybe some more information if there is a need. The second is in the process of being solved in proteomics.

In metabolomics they use the ISA-TAB format, and in transcriptomics the MAGE-TAB format is fulfilling a similar role. Proteomics has the SDRF format, which attempts to reach the same goal. Due to historical reasons it is not nearly as ingrained as the former two, it is also much younger than the other two and is therefore in large parts based on them. The goal is to create a format that will contain the necessary information to reproduce the data analysis. The general structure of the current version of this file splits the information into three groups, and pair them to cover the biological replicate times technical replicate characteristic of proteomics[38]. It is a vital foundation for the reanalysis of data, to compare them and make new findings.

This Study - NOT DONE YET

This project is a continuation of the study “Data mining antibody sequences for database searching in bottom-up proteomics” which was focused on the isotype distribution of B cells recorded in the Open Antibody Space. The original study was interested in the variable chain of the antibodies with the goal of creating an antibody database to support the discovery of unregistered antibodies[39]. Unlike the original this study is focused on the distribution of the variation in the Fc part of the antibody.

Method - NOT DONE YET

Data Collection

Data was gathered from the PRIDE repository. The goal was to gather datasets of serum samples, from as many different diseases as possible. First ChatGPT was asked for serum samples from various dis-

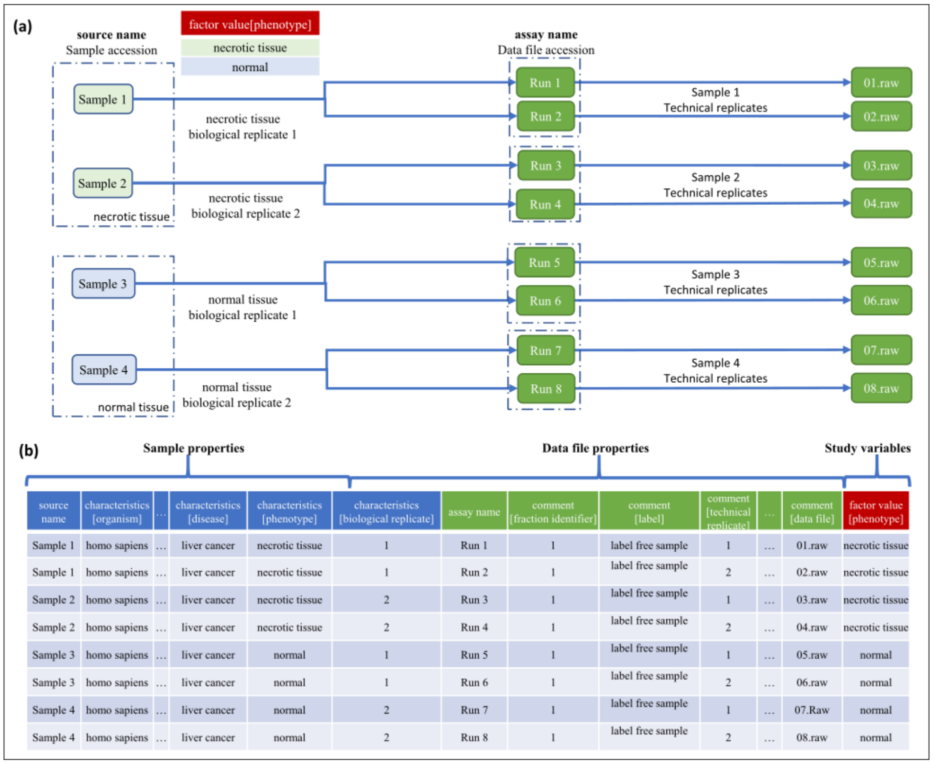

Figure 2: Structural overview of the SDRF format, Dat et al. 2021

eases, it forgot partway through the conversation that it was supposed to give serum samples, and it wasn't very thorough with the list it created. After having processed the datasets that the chatbot had found, another attempt was made with claude, which produced a python code for questioning the pride API, plus keywords. Finally a brute force search was conducted, systematically going through all entries on pride that had organism "homo sapiens" and organism part as one of the following; "blood serum, blood, or blood plasma. The datasets with promising names were written down in a document which can be found on the project github as https://github.com/Trenchcoat-capybara/Antibody-fingerprints/tree/main/pride_repository/chat_databases.md

The next step was to look through the article related to the datasets of interest to find the parameters to use for FragPipe analysis, and to see if there were any issues with the dataset in relation to the project itself. This aimed to remove the datasets where there were issue that made it useless for the study. The datasets where there was a need to guess which dataset belonged to control and which was of interest. The datasets where antibodies were removed fo increase the clarity of other proteins. The datasets that were targeting something even more specific in the blood or serum.

Data Treatment

After the datasets were verified, the few that had all the information needed, a proper separation of samples noted, without any antibodies removed, and not targeting something irrelevant in the blood, were ran through FragPipe, using the appropriate workflow, which leans on the DIA-NN software.

The chosen dataset was accession number PXD036074 which was decompressed with a dockerised version of proteowizard using the msconvert tool. The main tags of interest was peakPicking and compression. They were then transfered to the compute instance where they were analysed using 4 cores of with 6 gb of memory each.

The decoded data files were analysed in fragpipe using the DIA_SpecLib_Quant with the following parameters. The enzyme was set to trypsin. The precursor size was 12 ppm and the fragmentation size to 20 ppm. The instrument was SCIEX 7600 ZenoTOF in the acquisition mode Zeno Swath MS (DIA). It had the fixed modifications carbamidomethyl-Cys, and the variable mods were oxidation-Met and N-term Acetyl. It came from the organism Homo Sapien. The data was created with a label free DIA workflow.

The spectral libraries were aggregated (with csvstack from csvkit) the different patients into a single csv. After that the aggregated csv was inserted into postgres by using the csvsql function of csvkit. The files were then loaded using the DBI Rpackage, using the file names to sort the data according to disease and disease type.

The main data of interest in the output was the PG.Normalised values and their pairing to different isotypes. The PG.Normalised is the normalised registered quantity for the protein group. Each patient had several entries for each of the isotypes, so each of the isotypes was summed into a single value for each isotype per patient. In order to prepare the data for visualization, and to make it comparable across samples, the values were turned into percentages. They were then turned into two sets of data, one with all of the subtypes, and one with the subtypes summed up to represent the distribution between isotypes only.

Data Visualization and Statistics

Data visualization was performed with R using the ggplot2 package. The goal was originally to compare the distribution of isotypes accross different disease categories, but due to a lack of usable data some compromises had to be made, and only two disease which share a type were used. Statistical analysis was performed using the diseases as different groups, and on them after grouping by disease vector. The main statistical test was ANOVA as it was a comparison across groups.

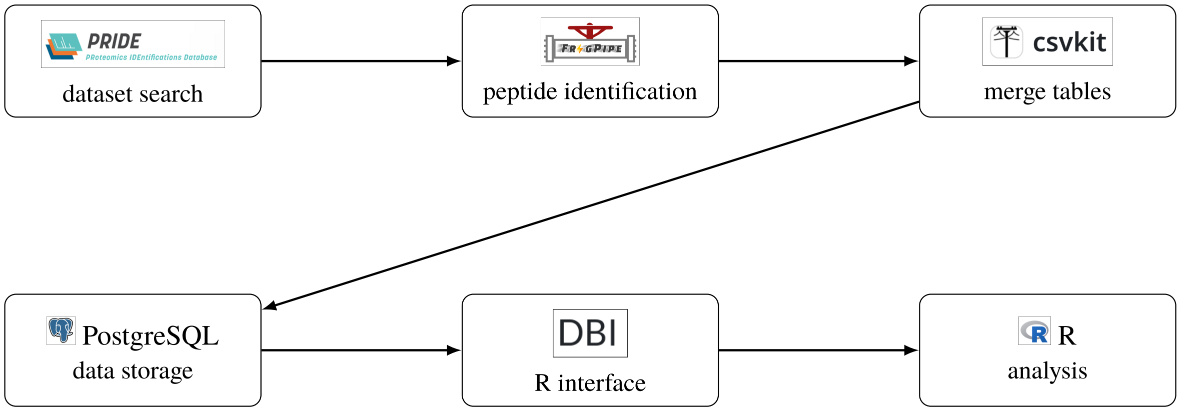

Figure 3: Overview of the workflow from dataset acquisition to statistical analysis.

Results

Distribution of the Data

The data that I was able to analyse was from the study "The human host response to monkeypox infection: a proteomic case series study". The study contained the following distribution of patients, and each patient had entries for the following unique isotype.

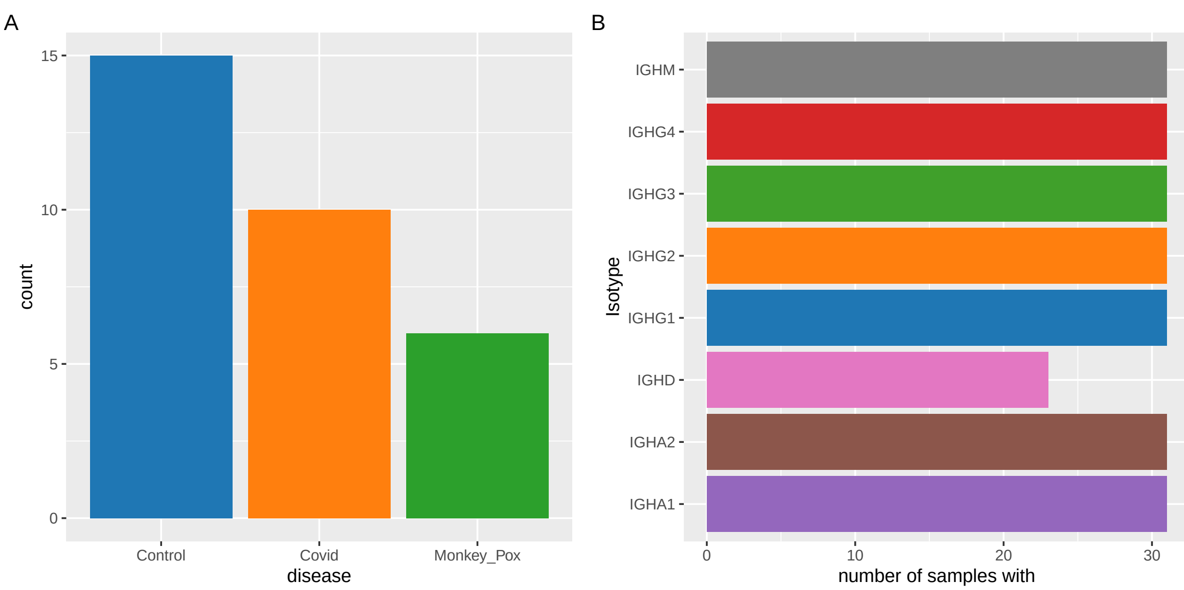

Figure 4: Distribution of disease and antibody isotypes across the datasets. (A) The number of patients of each type in the dataset. (B) The number of patients where the peptide was registered.

There are more control samples than there are samples for either of the disease, the two diseases combined contains one more sample than the control pool. It is interesting to note that there is not a single count of IgE, and the only isotype that is partially represented is IgD. The lack of IgE can be explained by the nature of IgE, but is there a reason for the missing counts of IgD?

IgE is the antibody with the lowest natural concentration in the blood. On top of that, it has the receptor with the highest affinity. That might be the reason that it wasn't present in a single sample, even though it was data collected with DIA methodology, which should be more sensitive to diverse low-quantity proteins. Another interesting reason may be that the patients in this data are from Britain. As IgE is a response to parasites, the historically improved sanitations - especially the separation of drinking water and excretion - has lowered the average level of IgE in the people of Britain, where the data is from.

IgD is generally produced in the peripheries which would mean that the source of the blood would have an effect on the concentration of IgD. Considering that the normal place to take blood from is the arm, which is far away from where IgD is produced, it may have an effect on the concentration of the IgD.

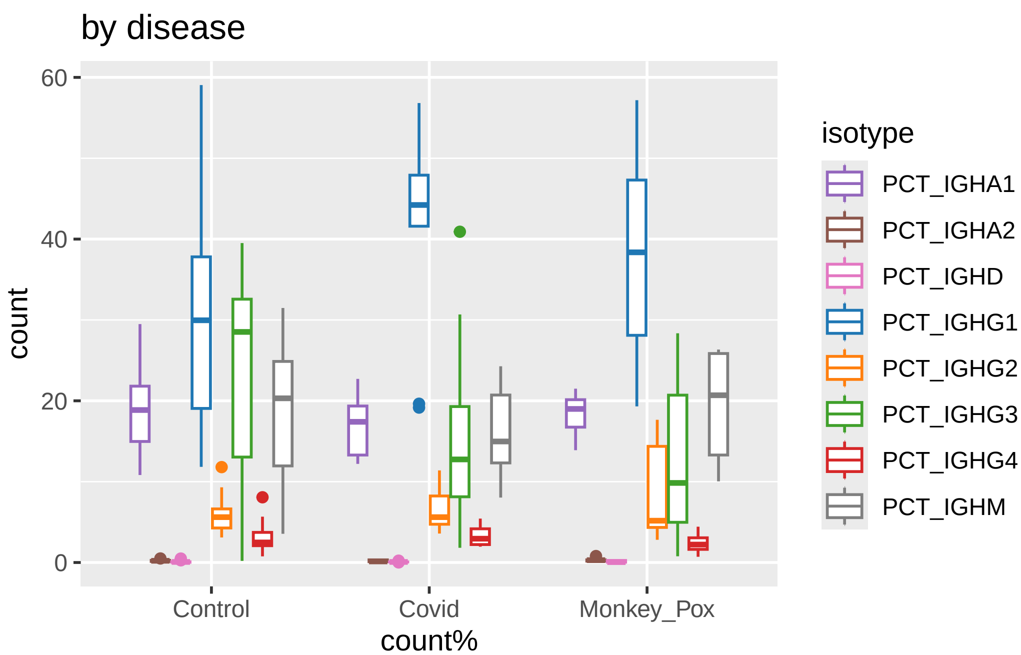

Figure 5: Distribution of isotypes for each disease.

As can be seen in Figure 5, there is a very low amount of IgD in the samples, this could be why there are so many IgDs that are not recorded, the amount may just stradle the line of detectability. This would overlap with the plausible reason that the IgE has not been found.

Another thing of interest is the Covid IgG1. Based on the boxplot shape, there seems to be 2 extreme outliers. This could be because there are only 10 samples in the study, meaning that it is easy for there to be a section where there weren't any reads, leading to them being classified as outliers. It would appear based on the graphs that the IgG1 outliers of Covid would fit in with the monkeypox values, this could make sense as both diseases are very similar, and should therefore bring forth a similar reaction from the immune system.

Comparing means across diseases

In order to get a better understanding of the differences in between the diseases and the control the means were calculated and plotted.

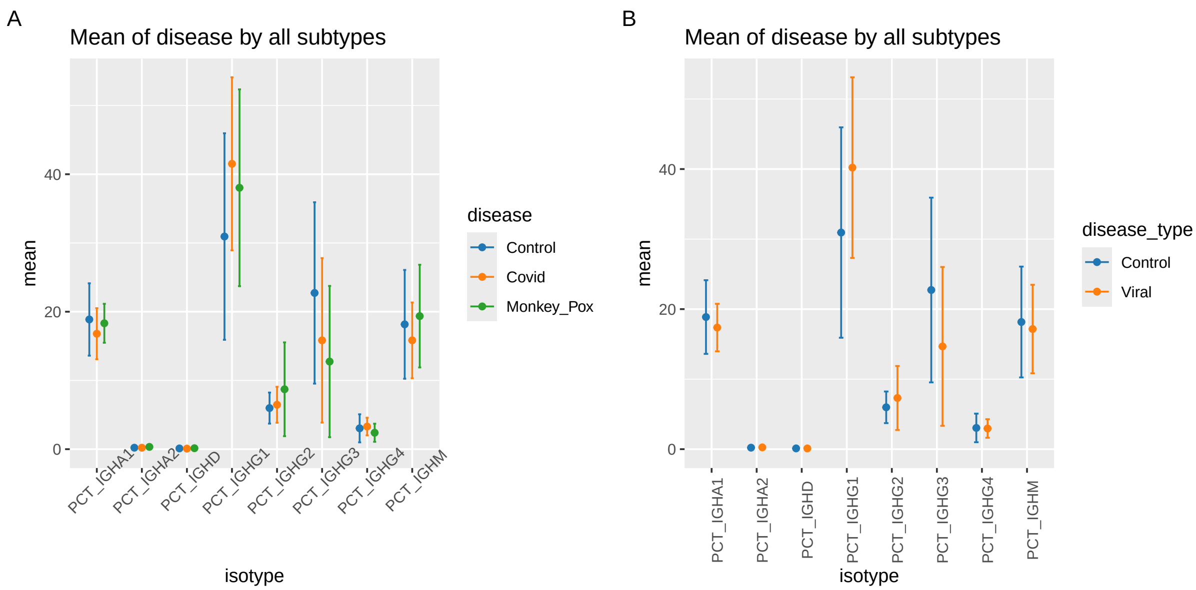

Figure 6: Comparison of mean percentage of antibody isotypes across for the three disease states.

With the overlapping differences in values between the control samples and the disease, there is a good chance that having the two disease separate will affect the statistical tests. This is predictable, as in the original study, they chose Covid as a comparison to their monkeypox cases due to the similarity between the two diseases. This could indicate that the idea to check isotype-fingerprints on a per disease type basis rather than on a per disease basis may be the more correct approach.

A quick look at the Isotypes show that the isotypes of most interest are IgG1 and IgG3, these are the only two where there appears to be a visible divergence between the control samples and the others. So statistical analysis will be performed on those two going forward. Due to the divergences being in IgG1 and IgG3, which are both IgG subtypes, it makes no sense to collapse the subtypes into their supertypes and analyse on that layer.

Analysis of vector

If the Covid and monkeypox are supposed to be treated as one group it should be relatively visible on a graph, In theory if they were one group then the samples should be distributed around a central mean in something approximating a normal distribution. As there are not a lot of samples there was very little to see, but there was once more a differerence from the control samples.

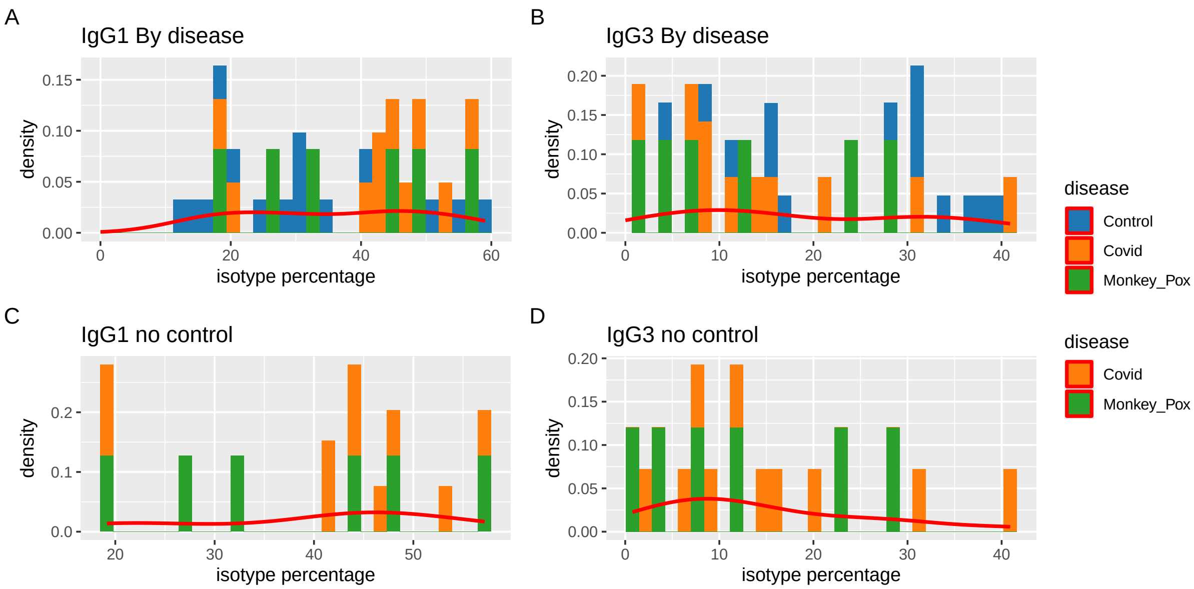

Figure 7: A collection of graphs visualising the distribution of patients percentages of the isotypes of interest. (A) and (B) are graphs containing all of the samples

There appear to somekind of relationship, but there are too few samples to say from a graph, it is therefore necessary to perform some statistical tests to see if the two viral samples should be counted separately. The control sample density grapths show two small bumps, this is another potential indicator that there are two sets here, Control and Viral.

Statistical Comparison of Viral Diseases

There are a couple of statistical tests that can be used to detect of there are differences between two groups. The most basic is to do perform an ANOVA test, comparing within group variation and between group variation.

```
     disease   isotype  n     mean      std      var
1    Control PCT_IGHG1 15 30.93733 15.02382 225.7151
2    Control PCT_IGHG3 15 22.73000 13.19194 174.0273
```

```
3      Covid PCT_IGHG1 10 41.51600 12.58922 158.4884
4      Covid PCT_IGHG3 10 15.83100 11.96752 143.2215
5 Monkey_Pox PCT_IGHG1  6 38.03000 14.31487 204.9154
6 Monkey_Pox PCT_IGHG3  6 12.74500 11.00355 121.0782
7      Viral PCT_IGHG1 16 40.20875 12.90102 166.4362
8      Viral PCT_IGHG3 16 14.67375 11.34342 128.6731
```

While the table shows that the means of the respective IgG1 and IgG3 concentrations of Covid and monkeypox are closer to each other than either of them are to the control sample, it doesn't really tell us if these are the same sample group or different.

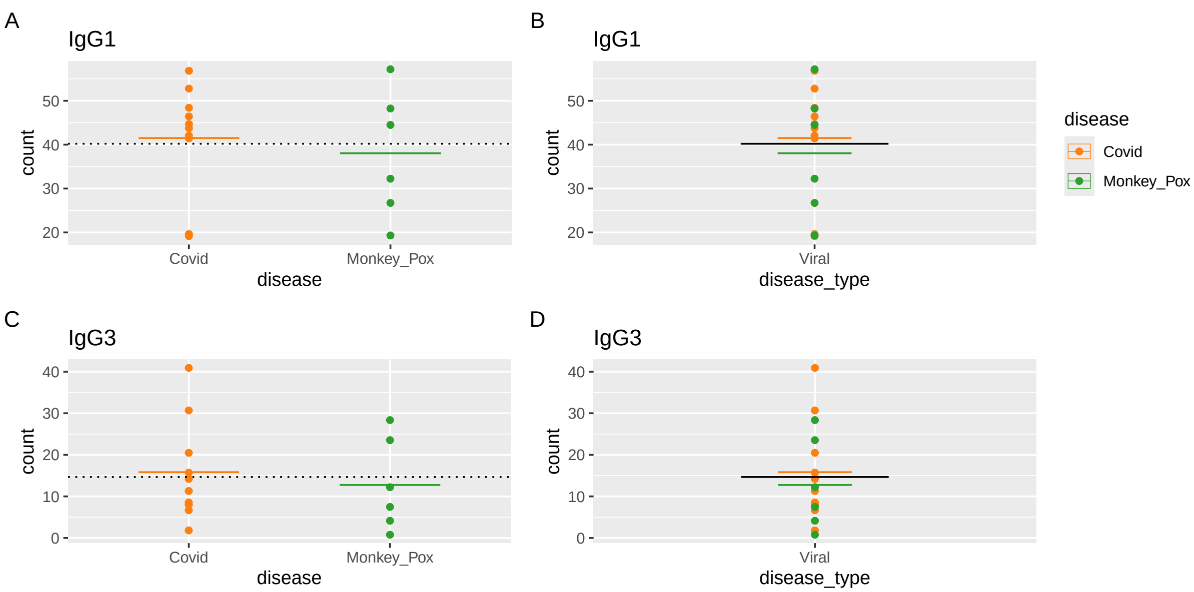

Figure 8: Visuallization of the overlap of covid and monkeypox datapoints. (A and C) is a comparison between the two diseases. (B and D) shows an overlap with a comparison of their individual means, and their collective mean. A B cover IgG1 and C D cover IgG3

As can be seen from the graphs there is a very high chance that the two groups are really one. An interesting aspect to this is that the two outliers from IgG1 Covid dataset are suddenly no longer outliers, they slot in perfectly with the viral disease type category. While this is a potential explanation, the low amount of samples makes it is hard to use this to prove that they are the same group.

As it cannot be shown that the two belong to the same group, comparison of groups with anova was first performed using the disease name as differentiating factor.

```
[1] summary of isotype ANOVA by disease for percentage of IgG1
```

```
Analysis of Variance Table

Response: table$count
              Df Sum Sq Mean Sq F value Pr(>F)
table$disease  2  711.1  355.53  1.7742 0.1882
Residuals     28 5611.0  200.39
```

```
[1] summary of isotype ANOVA by disease for percentage of IgG3
```

```
Analysis of Variance Table

Response: table$count
              Df Sum Sq Mean Sq F value Pr(>F)
table$disease  2  538.2  269.09  1.7398  0.194
Residuals     28 4330.8  154.67
```

With a Pr(>F) of around 0.19 for both isotypes it would inidicate that having a covid or monkeypox has no significant effect on the isotype-fingerprint of a person. Based on the earlier graphs this makes less sense than the idea that the double count of viral infection is interfering.

```
[1] "summary of isotype ANOVA by disease type percentage of IgG1"
```

```
Analysis of Variance Table

Response: table$count
                   Df Sum Sq Mean Sq F value  Pr(>F)
table$disease_type  1  665.5  665.49  3.4118 0.07496 .
Residuals          29 5656.6  195.05
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
[1] "summary of isotype ANOVA by disease type percentage of IgG3"
```

```
Analysis of Variance Table

Response: table$count
                   Df Sum Sq Mean Sq F value  Pr(>F)
table$disease_type  1  502.5  502.48  3.3372 0.07804 .
Residuals          29 4366.5  150.57
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

By counting the viral infections as a single group the Pr(>F) falls to around 0.075 which is not below the general significance level of 0.05. What it is, is within the range that usually inidicate weak correlation, the a value range of 0.10 to 0.05. Considering the lacking number of samples this is a none definite result and should probably be researched further. It does indicate that there is something there.

Discussion

Database sourcing with LLMs

In order to speed up the process of dataset discovery LLMs were used. The two used were ChatGPT and Claude Sonnet.

When ChatGPT was asked for datasets containig serum or blood samples for different diseases, it only gave a few even when asked for as many as possible. When asked for further datasets it seemed to 'forget' that the datasets had to be serum or blood samples.

Claude had a different approach, it said it couldn't access the PRIDE API and then produced a python script that could be used to access the API. The script wasn't completely thought through, as it was only able to do keyword search. This made it hard to specify it as only serum samples, and it was just easier to do a generalised keyword search using the PRIDE websites inbuilt search engine, even though it would be preferable to have AND/OR operators in that.

Exhaustion of Proteins

One of the common problems for finding datasets was the exhaustion of the most common proteins in the sample. These most common proteins usually include both IgG1 and IgA, which is understandable for analysis such as DDA but causes trouble for others when looking for that data, if it is not directly mentioned. While it is easy to comment that it exists in the method section of the report, when it is the umpteenth report of the day to find data it is not very helpful.

It could be intersting to add some fields to the properties tab in the PRIDE archive. Maybe something like a "kits used" or "removed proteins". This may be a bit overkill, or it maybe a bit vague. The main goal of this kind of implementation would be to make it easy to see at an instance that this is a dataset where, "this kind" of modification has been performed. It would spare people from digging into the report itself for that single line to see if the data is useful. They should obviously read the report if they intend to use the data, but that choice should idealy be made as simple as possible. By eliminating as much friction as possible it would be possible to further the use of the data as much as possible.

Annotation of Data

There was a severe lack of usable annotation of data on PRIDE. Several of the datasets were had files that were only differentiated by number, which makes the only way of knowing which are the control contacting the original makers of the data. It would appear that, to many getting the data onto PRIDE is an afterthought and no time has been put into thinking about what information needs to be included. This could be seen by the constant lack of things like the precursor size, and dataset labeling. It would be interesting if PRIDE added some datapoints that must be filled in to allow the datasets to be posted there. If they actually performed the experiment they should have notated the precursor size, fragmentation size, and other processing information somewhere and it would therefore be easy to make that available for others to redo the data analysis.

Once more it could be useful to have some more properties fields in the archive. It could be sub categories of the Experiment type, or software, the important part is to make it possible to reanalyse the data while lowering the amount of extra work that has to be done. The objective is once more to make it as painless as possible to reuse the data, thereby increasing the potential for future research. This was a non-exhaustive list of parameters, butthe same basic idea would be applicable here. These kinds of fields could also be enforced on an SDRF format.

Another project that could be of use, would be to have a team that goes through older datasets and with the help of the researchers create the SDRF files that are missing. These SDRF files would make the older datafiles that lack the information needed for reusability give the data new life.

Statistical testing - NOT DONE YET

There are several statistical limitations to the study, but the main is the lack of samples. This lack of samples makes any reached conclusion lack statistical proof of being correct.

Even though the reliability of the statistical testing is hindered by the low sample pool, there are results of potential interest. When initially treating the data the disease type was of secondary importance, and that led to results where the two viral vectors interfered with each other. With the visible difference in the values from statistical testing between using disease and factor versus using the disease vector as the factor. The p-value differences between the two sample groupings indicate the importance of grouping for future experiments.

While there was an intention to see if it was possible to use a random forest to find constant differences between different categories of disease. With the small sample size and disease type pool it makes no sense to fit a random forest on top of it.

The Importance of subtypes

With the OAS being isotype distribution data based on genetic data. It raises the question, is it necessary to include the subtypes? It is easier to perform an analysis using the genome of extracted cells. This could be an incentive, but that would presume that every B cell is producing the same amount of antibodies, and the subtypes are not of major importance. There is a known difference between the active production of antibodies of different types of B cell. Different types of antibodies produce different amounts of antibodies. But even if that is accounted for there seems to be a need to take account of the subtype as well. As can be seen from the data in Figure 6 the increase in IgG1 is similar to the decrease in IgG3, this difference would be lost when using genomics in order to record the change in isotype fingerprint.

It would also seem that there is no need to strictly classify the data entries according to individual diseases. This may be due to Covid and monkeypox being disease that are very similar in terms of function. It could be interesting to see if this would hold up with other viral diseases that function in a different way. With the current data there is an indication that the main factor of importance for the change in isotype fingerprint is the disease vectors type. Although this is hard to conclude on as Covid and monkeypox were chosen because they are very similar in several aspects.

Sample type

The samples being serum samples may skew the viewed isotype distributions. This probably has an greater effect on IgD and IgE as can be seen from the amount of samples that contain these two isotypes in Figure 4 which shows that 23 out of the 31 samples contain IgD and not a single sample registered IgE.

IgD is naturally concentrated around the peripheral areas of the circulatory system. It would therefore make sense that samples taken from other places would contain a much lower level of IgD. The lower levels will skew the isotype balance away from the IgD hidding anything that may be gained from knowing the true value.

IgE on the other hand is completely missing. The naturally low levels of IgE combined with it being concentrated on mast cells is probably the main culprit here. When combined with the fact that people living in countries with good sanitation, like Great Britain in the study, generally are less exposed to the parasites that trigger the production of IgE. It creates a scenario where people have low levels of IgE that get snatched up by the mast cells. So for any information from IgE is completely lost.

This may be mitigated by taking serum samples from less intuitive places, and if it shows that those serum samples differ from the standard serum samples. This thought can be extended to IgA as most of that is secreted, maybe the comparison between the IgA subtype levels in certain secretions could bare more information for analysis, although considering that IgA is closer to a preventative subtype this feels redundant. What could be more interesting is the usage of alternative serum sampeling areas fro different things, it might not only be antibodies that are non-equally distributed throughout the body.

Conclusion

This study attmpted to use open source data from past experiments to show that there was a relationship between disease and antibody isotype fingerprint. It uses data from the PRIDE repository in order to test if there is any visible statistical difference in the isotype distribution when afflicted with monkeypox, covid, or not afflicted at all. It would appear that the isotype fingerprint of a person changes while they have an affliction, it would also appear that it is based on the vector type or attact pattern rather than varying from individual disease to disease. There is not enough evidence to prove it definitely in this data. Future experiments could verify the finding, and attempt to expand on it for other disease categories as well as see if there is more variation to be found than simply the identity of the vector. Another area that definately requires future work done, would be in fleshing out the data parameters on PRIDE, this would make the data there actually useful for future studies and reviews, instead of just creating the illusion of usefulness.

Data

The source code for this project can be found on github at https://github.com/Trenchcoat-capybara/Antibody-fingerprints.git. This is a pdf rendering of the quarto document main.qmd which can be found in the root of the project.

Run	id	disease	accession_number
20220624_Z1_ZW_001_30-0066_Ctr-A10_heat_DIA	1	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A11_heat_DIA	2	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A12_heat_DIA	3	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A1_heat_DIA	4	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A2_heat_DIA	5	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A3_heat_DIA	6	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A4_heat_DIA	7	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A5_heat_DIA	8	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A6_heat_DIA	9	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A7_heat_DIA	10	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A8_heat_DIA	11	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-A9_heat_DIA	12	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-B1_heat_DIA	13	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-B2_heat_DIA	14	Control	PXD036074
20220624_Z1_ZW_001_30-0066_Ctr-B3_heat_DIA	15	Control	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E10_heat_DIA	16	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E1_heat_DIA	17	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E2_heat_DIA	18	Covid	PXD036074

20220624_Z1_ZW_001_30-0066_COVID-E3_heat_DIA	19	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E4_heat_DIA	20	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E5_heat_DIA	21	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E6_heat_DIA	22	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E7_heat_DIA	23	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E8_heat_DIA	24	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_COVID-E9_heat_DIA	25	Covid	PXD036074
20220624_Z1_ZW_001_30-0066_MPX-D1_heat_DIA	26	Monkey_Pox	PXD036074
20220624_Z1_ZW_001_30-0066_MPX-D2_heat_DIA	27	Monkey_Pox	PXD036074
20220624_Z1_ZW_001_30-0066_MPX-D3_heat_DIA	28	Monkey_Pox	PXD036074
20220624_Z1_ZW_001_30-0066_MPX-D4_heat_DIA	29	Monkey_Pox	PXD036074
20220624_Z1_ZW_001_30-0066_MPX-D5_heat_DIA	30	Monkey_Pox	PXD036074
20220624_Z1_ZW_001_30-0066_MPX-D6_heat_DIA	31	Monkey_Pox	PXD036074

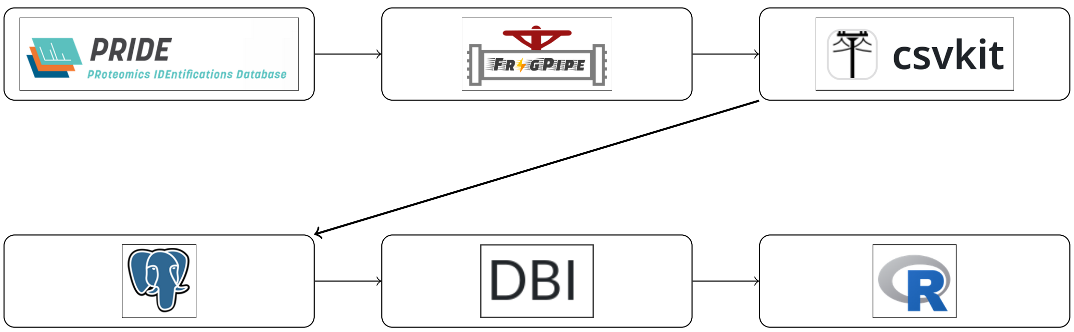

Figure 9: Overview of the workflow. Start by searching PRIDE for datasets. Run the datasets through FragPipe. Use csvkit to merge the outputs into a single table that can be used, and put into postgres. Use Postgres for data management and DBI to interact with it in R. Do the statistical work in R.

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